

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Roomly - Smart Property & Tenant management Platform

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Roomly - Smart Property & Tenant management Platform**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Shikha Tuteja. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date: 29-April-2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Introduction

1.1 Problem Background

Traditional rental management systems rely heavily on manual processes such as maintaining registers, spreadsheets, and offline communication. These methods lead to errors in billing, lack of transparency, inefficient tenant management, and difficulty in handling multiple properties. There is no centralized platform that provides all services in one place.

1.2 Objective of the Project

The main objectives of Roomly are:

- To develop a centralized property management system
- To automate billing and payment tracking
- To manage tenants and properties efficiently
- To provide a scalable and user-friendly platform
- To support dynamic user roles (landlord and renter)

2. METHODOLOGY

The project follows a modular and structured development approach:

- Requirement analysis and problem understanding
- Designing system architecture and database schema
- Backend development using Node.js and Express.js
- Frontend development using React.js
- Integration of modules and API testing
- Deployment and testing

3. TOOLS AND TECHNOLOGIES

The following technologies are used:

- Frontend: React.js
- Backend: Node.js, Express.js
- Database: MongoDB
- Authentication: JWT (JSON Web Token)

- Version Control: Git & GitHub
- IDE: VS Code

4. IMPLEMENTATION

The Roomly system is implemented using a full-stack architecture. The backend handles business logic, authentication, and database operations, while the frontend provides a user-friendly interface.

Key modules implemented:

- User Authentication System
- Property Management (House, Floor, Room)
- Billing System (Rent + Electricity + Charges)
- Payment Tracking
- Maintenance Request Module
- Notification System

5. MAJOR FINDINGS / RESULTS

The developed system successfully:

- Automates rental management processes
- Reduces manual errors in billing
- Improves communication between landlord and tenant
- Provides real-time updates and notifications
- Supports multiple properties with structured hierarchy

The system was tested and found to be efficient, scalable, and user-friendly.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

Roomly provides an effective solution for modern property management by integrating multiple functionalities into a single platform. It simplifies complex processes and enhances efficiency for both landlords and tenants.

6.2 Future Scope

- Mobile application development
- AI-based analytics and insights
- Real-time chat system
- Online payment gateway integration
- Advanced reporting dashboard

REFERENCES

- React.js Documentation – <https://reactjs.org/>
- Node.js Documentation – <https://nodejs.org/>
- Express.js – <https://expressjs.com/>
- MongoDB – <https://mongodb.com/>
- JWT – <https://jwt.io/>

APPENDICES

- Project Screenshots
- System Architecture Diagram
- User Manual

